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TITLE

Prediction of structure, finding critical residue and interacting sites of antitoxin toxin pairs of *Neisseria Lactamica*, and study dynamics of proteins using Molecular Dynamics Simulation

Project report submitted in partial fulfillment of the requirements for the award of the degree of

Masters of Technology
Biotechnology & Biochemical Engineering

By

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Under the supervision of
Prof. Amit Kumar Das
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Department of Biotechnology
Indian Institute of Technology Kharagpur

CERTIFICATE

This is to certify that the project report entitled "**Prediction of structure, finding critical residue and interacting sites of antitoxin toxin pairs of *Neisseria Lactamica*, and study dynamics of proteins using Molecular Dynamics Simulation**" is submitted by **Mahesh Jethalia (15BT30036)** to the Department of Biotechnology, in partial fulfillment for the award of the degree of Bachelor of Technology, is an authentic record of the work carried out by him under my supervision and guidance. The report has fulfilled all the requirements as per the regulations of this institute.

Date: May 26, 2020

Prof. Amit Kumar Das

Department of Biotechnology

Indian Institute of Technology Kharagpur

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Date: 5th June 2020 Mahesh Jethalia

Place: Kharagpur Department of Biotechnology

Abstract

Type II toxin-antitoxin systems are generally better-understood than type I. In this system a labile protein antitoxin tightly binds and inhibits the activity of a stable toxin. The largest family of type II toxin-antitoxin systems is vapBC which has been found through bioinformatics searches to represent between 37 and 42% of all predicted type II loci.

Type II systems are organized in operons with the antitoxin protein typically being located upstream of the toxin, which helps to prevent the expression of the toxin without the antitoxin. The proteins are typically around 100 amino acids in length and exhibit toxicity in a number of ways.

Introduction

Fast intracellular trafficking (fitAB) locus cause the bacteria to transit a polarized epithelial monolayer more quickly than the wild-type parent and to replicate within cells at an accelerated rate. Here, we describe the crystal structure of the toxin-antitoxin heterodimer, FitAB, bound to a high affinity 36-bp DNA fragment from the fitAB promoter. FitA, the antitoxin, binds DNA through its ribbon-helix-helix motif and is tethered to FitB, the toxin, to form a heterodimer by the insertion of a four-turn α-helix into an extensive FitB hydrophobic pocket. FitB is composed of a PIN (PilT N terminus) domain, with a central, twisted, 5-stranded parallel β-sheet that is open on one side and flanked by five α-helices. FitB in the context of the FitAB complex does not display nuclease activity against tested PIN substrates. The FitAB complex points to the mechanism by which antitoxins with RHH motifs can block the activity of toxins with PIN domains. Interactions between two FitB molecules result in the formation of a tetramer of FitAB heterodimers, which binds to the 36-bp DNA fragment and provides an explanation for how FitB enhances the DNA binding affinity of FitA.

Molecular dynamics simulation, i.e., preparation, minimization, and equilibration of your system typical simulation techniques and the analysis of equilibrium properties.

Theory

Mimic what atoms do in real life, assuming a given potential energy function – The energy function allows us to calculate the force experienced by any atom given the positions of the other atoms – Newton's laws tell us how those forces will affect the motions of the atoms.

Molecular Dynamics simulations can be used in determining where drug molecules bind, and how they exert their effects We used simulations to determine where this molecule binds to its receptor, and how it changes the binding strength of molecules that bind elsewhere (in part by changing the protein's structure). We then used that information to alter the molecule such that it has a different effect.

Determining functional mechanisms of proteins. Molecular Dynamics Simulations can be used to describe the mechanism by which drugs binding to one end of the receptor cause the other end of the receptor to change shape (activate). It can be used for studying protein folding.

Molecular dynamics simulations permit the study of complex, dynamic processes that occur in biological systems. These include, for example,

- Protein stability
- Conformational changes
- Protein folding
- Molecular recognition: proteins, DNA, membranes, complexes
- Ion transport in biological systems

and provide the mean to carry out the following studies,

- Drug Design
- Structure determination

Methods

Steps of MD Simulation in NAMD:
Download and Extract the namd_tutorail.zip file and extract it in Desktop.
A. PDB File Processing
1) First, you will remove the water molecules from the pdb file, and create a pdb file of the protein alone.
2) Open VMD and load the pdb file.
3) From Graphics -> Representations create a new representation for ‘protein’ selection.
4) Select the molecule in the VMD Main window. Go to File -> Save Coordinates.
5) In the Selected Atoms of the Save Trajectory window select ‘protein’
6) Click on ‘Save’
7) Save the file with the name ‘****.pdb’ in the namd folder in your Desktop.
Note that the X-ray structure from the Protein Data Bank does not contain the hydrogen atoms of ubiquitin. This is because X-ray crystallography usually cannot resolve hydrogen atoms. The pdb file you will generate with psfgen along with the psf will contain guessed coordinates for hydrogen atoms of the structure. Later, the energy minimization of the protein will ensure their positions are reasonable.
B. Generating a Protein Structure File (PSF)
1) Now, you will create the psf file of ubiquitin. First load the ubprotein.pdb in the VMD Main. VMD offers an automatic psf file builder via the VMD Main menu by clicking Extensions -> Modeling -> Automatic PSF Builder.
2) Now in Automatic PSF Builder Click on Load Input Files and Select the Protein for PSF
preparation. Click on Guess and Split Chains using current selections. Finally, generate the
psf file by clicking on I am feeling lucky.
3) This process will generate two files ubprotein_autopsf.psf and ubprotein_autopsf.pdb
which can be found in C:\-> Users -> <User Name>-> AppData -> Local -> VirtualStore -
> Program Files (x86) -> University of Illinois -> VMD
4) Copy these files to ‘namd’ directory in Desktop
So, you have created the psf! You may want to inspect ubprotein_autopsf.psf and

ubprotein_autopsf.pdb using WordPad.

C. Solvating the Protein

Now, the protein needs to be solvated, i.e., put inside water, to more closely resemble the cellular environment. You will do so in two ways, placing ubiquitin in a water box, in preparation for minimization and equilibration with periodic boundary conditions.

1) Form VMD Main window goes to Extension -> TkConsole window.

2) Change your Working directory to the namd directory in your Desktop using the following command.

% cd C:/Users/<user name>/Desktop/namd/

3) Now solvate the protein with the following command:

% solvate ubprotein_autopsf.psf ubprotein_autopsf.pdb -t 5 -o water

The solvate command loads the solvate package so that VMD will be able to call it. The solvate package will put your protein (described in ubprotein_autopsf.psf and ubprotein_autopsf.pdb) in a box of water. The -t option creates the water box dimensions such that there is a layer of water 5Å in each direction from the atom with the largest coordinate in that direction. The -o option creates the output files water.pdb and water.psf for ubiquitin with the water box.

Now, you can load the water.psf and water.pdb to visualize the water box surrounding the protein.

D. Adding Ions to the system

Ions should be placed in the water to represent a more typical biological environment. They are especially necessary if the protein being studied carries an excess charge. In that case, the number of ions should be chosen to make the system neutral. The ions present will shield the regions of the protein which carry the charge and make the entire system more stable. They should be placed in regions of potential minima since they will be forced to those regions during the simulation anyway. The psf file contains the charge of each atom and may be used to determine the charge of the total system or parts of it.

We will add 150mM of NaCl to the system. In the VMD TkConsole Window enter the following command:

% autoionize -psf water.psf -pdb water.pdb -sc 0.15 -cation SOD -anion CLA -o ionized

This command will call the autoionize plugin. It will neutralize the system first and add an additional 150mM NaCl. You will find two newly generated files i.e. ionized.psf and ionized.pdb

E. Ubiquitin is a Water Box: Simulation with Periodic Boundary Conditions

Open the configuration file, ub_min_eq.conf in your namd directory. The configuration file may seem complex at first, but it will be examined line by line to determine its function in your simulation. Note that when "#" appears at the beginning of a line, the entire line is treated as a comment and ignored by NAMD. Detailed information of Configuration File Formats are given in the "Introduction to File Formats.pdf" provided with this tutorial. Though some basic features are explained here.

1. The "Job Description" section contains only comments, and its purpose is to inform those who view the configuration file about what it is meant for. Your comment should read :

Minimization and Equilibration of
Ubiquitin in a Water Sphere

2. The "Adjustable Parameters" section contains five commands:

structure: calls the psf file describing the system (ionized.psf)

coordinates: calls the initial coordinate data from the file listed next to it (ionized.pdb).

set temperature: creates a variable called "temperature" in which to store a value for the initial temperature of the system. If you place the text $temperature in the configuration file, NAMD simply reads it as a label for the number "310". (Creating variables is useful since you can alter the value of that variable in a single place if it is listed many times in the configuration file.)

set output name: creates a variable called "output name" in which to store a generic name for output files. If you place the text "$output name" in the configuration file, NAMD simply reads it as a label for "ubqw_min_eq".

first timestep: simply sets a number value for the first time step of the simulation. It is typically useful when restarting a simulation. For instance, if a previous simulation ended at time step 552, the command firsttimestep 553 would be used.

3. The "Simulation Parameter" section contains many commands, commented into different categories:

Input

– paraTypeCharmm: indicates whether or not the parameter file is in the format used by the CHARMM force field. indicates that it is; off indicates that it is not. (If this command is not
specified, NAMD assumes the file is in X-PLOR format by default.)

– parameters: calls the force field parameters from the file listed next to it.

– temperature: sets the initial temperature of the system in Kelvin (K), with the value listed next to it (in this case $temperature, or 310). This is done by assigning random velocities to atoms picked from a Maxwell distribution such that their average kinetic energy accurately represents the given temperature.

Force-Field Parameters

– exclude: specifies which atomic interactions are to be excluded from consideration. See Figure 2 for general atom labels. The scaled 1-4 value indicates that interactions between any such atoms 1 and 2 and 1 and 3 are neglected and interactions between atoms 1 and 4 are weakened. The van der Waals interaction for “1-4" atoms are modified using special 1-4 parameters defined in the parameter files, and electrostatic interaction is modified as shown in the next command.

Fig. 2 Number labels for atoms which are a given amount of bonds away from one another

– 1-4scaling: specifies the degree to which the electrostatic interaction between 1-4 atoms is to be taken into account. It may be a decimal between 0 and 1 (here, it is 1) and indicates how much the interaction is “turned off" or “on", respectively.

– cutoff: indicates the distance in Å beyond which electrostatic and van der Waals interactions are cut-off. Otherwise, those interactions are considered over the entire volume of your system. This can be computationally too costly. When also employing a fast solver for electrostatics such as Particle Mesh Ewald or Multilevel Summation Method, the cutoff parameter instead defines the splitting distance for the 1= r interaction potential, in which a short-ranged part is evaluated exactly between atoms within the cutoff distance, and the remaining long-ranged part is approximated by the solver.

– switching: indicates whether or not switching functions are used to smoothly take electrostatic and van der Waals interactions to zero at the cutoff distance. You may list “on" or “off" next to it as a yes/no answer.

– switchdist: indicates the distance in Å at which the functional form of electrostatic and van der Waals interactions is modified to allow their values to approach zero at the cutoff distance. A visual explanation of this is useful and may be found in Figure 3.

– pairlistdist: is designed to make computation faster. It specifies a distance in Å. NAMD will only search within this distance for atoms which may interact by electrostatic or van der Waals interactions. This way, NAMD does not have to search the entire system. The distance must be greater than the cutoff distance, and the list must be updated during the simulation.

Integrator Parameters

– timestep: indicates the value of the time step size used in the simulation. MD simulations solve Newton’s laws in a discrete approximation to determine the trajectories of atoms. The time step tells NAMD how to discretize the particle

dynamics. It is specified in femtoseconds (here, 2 fs).
– rigidBonds: specifies which bonds involving hydrogen are considered to be rigid (non-vibrating). The value all specifies all linear bonds involving hydrogen and any other atoms.
– nonbondedFreq: specifies in the number of time steps how often nonbonded interactions should be calculated. It is useful for saving computational time.
– fullElectFrequency: specifies in a number of time steps how often full electrostatic interactions should be calculated.
– stepspercycle: Atoms are reassigned pair list identities (as explained above) once every cycle. This command specifies how long one cycle lasts, i.e. the number of time steps in one cycle.
Constant Temperature Control
– Langevin: indicates whether or not the simulation uses Langevin dynamics; uses values on and off. See the science box below for more on Langevin dynamics.
– langevinDamping: sets the value of the Langevin coupling coefficient, which quantifies the friction applied to the system, removing energy from the system, slowing atoms down, etc. It is specified in ps-1. langevinTemp: Langevin dynamics may be applied to all atoms or only non-hydrogen atoms in the system. This command specifies the temperature at which to keep those atoms, even though friction and random forces will be acting on them. (Remember $temperature is a variable for the value 310.) langevinHydrogen: indicates whether or not Langevin dynamics will be applied to hydrogen atoms in the simulation; uses values on and off.
Periodic Boundary Conditions
– Three periodic cell basis vectors are to be specified to give the periodic cell its shape and size. They are cellBasisVector1, cellBasisVector2, and cellBasisVector3. In this file, each vector is perpendicular to the other two, as indicated by a single x, y, or z value being specified by each. For instance, cellBasisVector1 is x = 42Å, y = 0Å, z = 0Å. With each vector perpendicular, a rectangular 3-D box is formed.
– cell origin: specifies the coordinates of the center of the periodic cell in Å. This is calculated after the solvation step!
– wrap water: This command may be used with periodic boundary conditions. If a water molecule crosses the periodic boundary, leaving the cell, setting this command to on will translate its coordinates to the mirror point on the opposite side of the cell. Nothing can escape. The command may be set to on or off.
– wrapAll: same as wrap water, except this applies to all molecules.
Electrostatics with PME.
Particle Mesh Ewald (PME) is a useful method for dealing with electrostatic

interactions in a system when periodic boundary conditions are present. The Ewald sum is an efficient way of calculating long-range forces in a periodic system. The particle mesh is a 3-D grid created in the system over which the system charge is distributed. From this charge, potentials and forces on atoms in the system are determined. As a result, your grid size should be chosen such that
it is fine enough to accurately represent the configuration of your system.
– PME: indicates whether or not the simulation uses the Particle Mesh Ewald Sum method; uses values yes and no.
– PMEGridSpacing: sets the minimum ratio between the number of PME grid points along each cellBasisVector and the physical dimensions. Since the grid will replicate the charge distribution in your system, PMEGridSpacing should be chosen to be large enough so that the grid spacing accurately reproduces your charge distribution. However, it should not be so large that it will slow down your simulation to provide unnecessary precision. Typically, a grid density of slightly more than 1/Å is a good choice to reproduce charge distribution in biological systems, where the closest atoms have a bond separation on the order of 1 Å. This corresponds to a PMEGridSpacing of 1.0. NAMD will then automatically set the PME grid sizes (see below) such that there is always less than 1.0 Å between grid points and the sizes contain only small prime factors (e.g. 2, 3, and 5).
Alternatively, one can define the PME grid sizes manually, using PMEGridSizeX, PMEGridSizeY, and PMEGridSizeZ. These set the size of the PME grid along cellBasisVector1, 2, and 3, respectively (not the x, y, and z directions as implied). For speed in computing Fast Fourier Transforms, PMEGridSizeX should be chosen so that it can be factored by 2, 3, or 5. If your cellBasisVector1 = (60, 0, 0), a good choice for PMEGridSizeX might be 64, since 60 Å / 64 = 0.9375 Å and 64 = 26. Note that since cellBasisVector is defined with slightly different values in each direction, the size of the mesh spacing (in length) will be different in each direction. Note also that when using the PME method, the command cutoff dictates the separation between long and short-range forces for the method; it does not simply turn off interactions.
Constant Pressure Control (variable volume)
– useGroupPressure: NAMD calculates system pressure based on the forces between atoms and their kinetic energies. This command specifies whether interactions involving hydrogen should be counted for all hydrogen atoms or simply between groups of hydrogen atoms; uses values yes and no and must be set to yes if rigidBonds are set.
– useFlexibleCell: specifies whether or not you want to allow the three dimensions of the periodic cell to vary independently; uses values yes or no.
– useConstantArea: NAMD allows you to keep the x - y cross-sectional area

constant while varying the z dimension; uses values yes and no. Langevin piston: indicates whether or not the simulation uses a Langevin piston to control the system pressure; uses values on and off. Langevin pistonTarget: specifies, in units of bar, the pressure which the Langevin piston tries to maintain. (1 atm = 1.013 bar)
The following are specifications for the Langevin piston which NAMD allows you to specify. langevinPistonPeriod sets the oscillation time constant in fs for the Langevin piston. langevinPistonDecay sets the damping time constant in fs for the Langevin piston. langevinPistonTemp: sets the “noise" temperature in K for the Langevin piston; it should be set equal to the target temperature for the temperature control method (here, set by langevinTemp).
Output
– outputName: Several types of output data may be written by NAMD for any simulation. This command specifies the prefix for output filenames. NAMD always returns the following two output files for every simulation: a pdb file containing the final coordinates of all atoms in the system and a pdb file containing the final velocities of all atoms in the system; the extensions for these files are .coor and .vel, respectively. Thus, this configuration file will write two files named ubq_min_eq.coor and ubq_min_eq.vel
– restartfreq: During the simulation, NAMD can also create restart files, one of which is a pdb file which stores atomic coordinates, and the other of which stores atomic velocities. This
command specifies the amount of time steps between writing to the restart file (here, every 500 steps, or 1000fs, or 1 ps). If this command is not set, NAMD will not create a restart file. Furthermore, NAMD will store the file from the previous cycle each time it writes a new file. The filename is appended with a .old extension; it is created in case NAMD fails in writing the new restart file.
– dcdfreq: The dcd file contains only atomic coordinates, and they are written to the file several times over the course of a simulation. Thus, it provides a trajectory of the system over the runtime. This command specifies the number of time steps. between writing new coordinates to the dcd output file. If this command is not set, NAMD will not create a dcd file. In addition to the output files described, NAMD also prints a log of the simulation (which we will redirect into a .log file). This output is explained further below.
– output energies: specifies the number of time steps between each output of system energies (for various force field interactions) into the .log file (here, every 100 steps, or 200 fs).
– xstFreq: The extended system trajectory file contains a record of the periodic cell parameters, essentially recording the trajectory of the cell boundaries over the run

time. This command
specifies how often, in time steps, the configuration will be recorded. If this command is set, three xst files will be output: 1 final and 2 restarts.
– output pressure: specifies the number of time steps between each output of system pressure into the .log file.
4. The “Execution Script" section contains three commands, the first two of which apply to minimization and the last one of which applies to equilibration.
Minimization
– minimize: sets the number of iterations over which to vary atom positions to search for a local minimum in the potential (in this case 100).
– reinitvels: Minimization is performed on the system after all atomic velocities have been set to zero. This command resets atomic velocities such that the system starts at the temperature
specified (in this case, $temperature, or 310K).
Run: sets the number of time steps over which to run the MD equilibration (in this case 2500, which corresponds to 5,000 fs or 5 ps, since a 2 fs time step has been used).
Running the Simulation:
1) Now, in windows search type cmd for Command Prompt. By default, it will open
C:\Users\<user name>>
Go to the ‘namd’ directory on the desktop.
C:\Users\<user name>> cd Desktop\namd
2) Now, Run the simulation with the ub_min_eq.conf file provided by the command below:
C:\Users\<user name>\ Desktop\namd>NAMD_2.12\namd2 ub_min_eq.conf
You will see the process running in the cmd window. You can save the output of this process to a log file with the command below:
C:\Users\<user name>\ Desktop\namd>NAMD_2.12\namd2 ub_min_eq.conf > ub_min_eq.log
4.3 Analysis of the Simulation:
So, we have performed the energy minimization and equilibration of the ubiquitin in water. Load the water.psf and the ubq_min_eq.dcd in VMD to visualize the movements of the protein throughout the process. Now we will use the VMD RMSD Trajectory tool and HBonds Plugin (from VMD Main -> Extension -> Analysis) for some basic analysis.
A. RMSDTT: RMSD Trajectory Tool
RMSD measures the deviation of a target set of coordinates (i.e. a structure) to a reference set of coordinates, with RMSD=0.0 indicating a perfect overlap. RMSD

is defined as:
Where N is the number of atoms, mi is the mass of atom I, Xi is the coordinate vector for target atom I, Yi is the coordinate vector for reference atom I, and M is the total mass. If the RMSD is not mass-weighted, all mi = 1 and M = N.
The basic features of the tool are:
Calculate rmsd in trajectories, skip frames, and do an all-to-all rmsd (all molecules/frames against each other).
Calculate weighted rmsd. Align frames in trajectories. Iterative fitting.
Plot results via the Multiplot VMD plugin (in all systems), Xmgrace (on Unix systems), or MS Excel (on windows). Save results to file.
Swap equivalent atoms (i.e. Glu OE1 and OE2) to reduce rmsd (if Swap plugin is available).
Generate basic statistics (average, standard deviation, minima and maxima).
Multi-line atoms selection with support for comments.
Highlight equivalent atoms of the rmsd selection.
B. HBonds Plugin
The H-bonds plugin counts the number of hydrogen bonds formed throughout a trajectory. The search can be restricted to a single selection or between two distinct selections, as well as a frame range given by the user.
Criteria for the formation of a hydrogen bond: A hydrogen bond is formed between an atom with a hydrogen-bonded to it (the donor, D) and another atom (the acceptor, A) provided that the distance D-A is less than the cut-off distance (default 3.0 Angstroms) and the angle D-H-A is less than the cut-off angle.

Analysis

BLAST and BLASTp were used to find related sequences. CLUSTALW was used to find the conserved regions across different strains of bacteria for both the toxin and antitoxin.
A small program written in C was used to find interacting residues with distance among them to be less than 4.5 Armstrong. Swiss-Model was used to undergo Homology Modelling and PatchDock was used to generate the docked structure of the protein.

BLAST was used on the toxin and antitoxin strains of Neisseria Lactamica which resulted in us selecting FitA and FitB. The structure of the proteins of Neisseria was not available on RCSB PDB. Hence we used Homology Modelling using SWISS-MODEL to generate a model for both toxin and antitoxin of Neisseria.

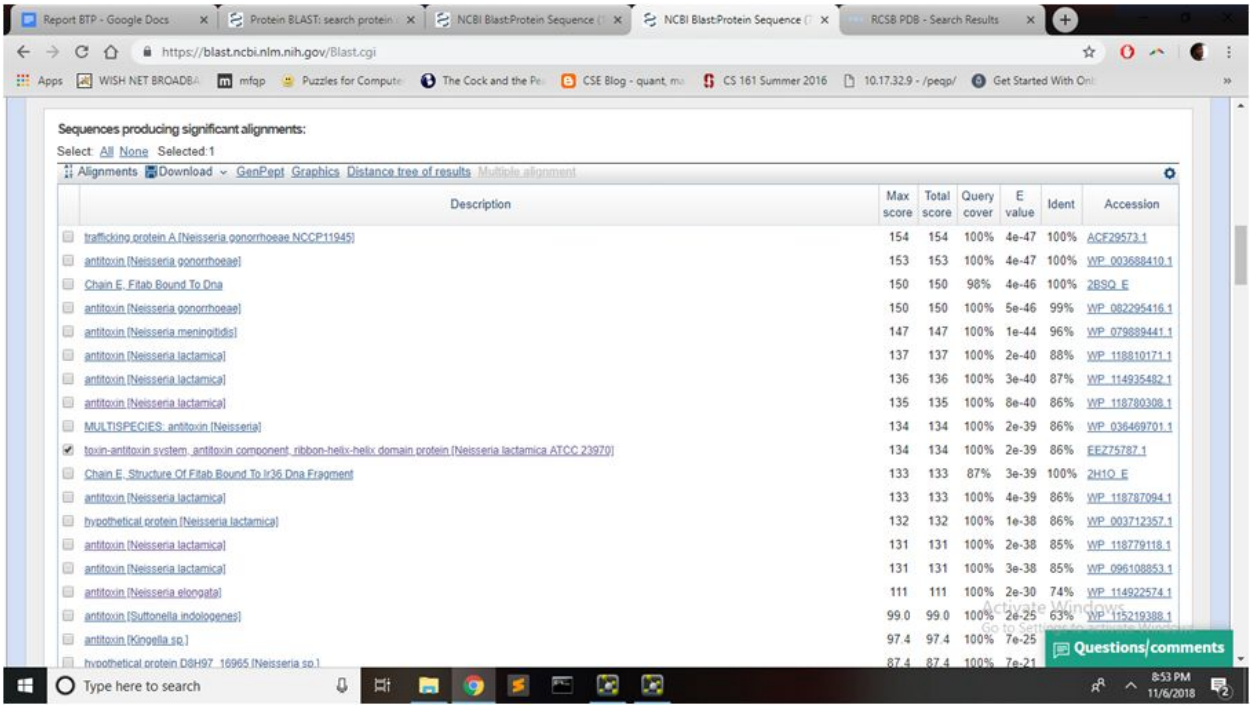


Figure 1 : BLAST

SWISS-MODEL used the template of FitA and FitB to generate the homology models. Using that we used PatchDock to generate the docked proteins. To find interacting sequences we used a small C code and used distance to be less than 4.5 Armstrong. We got 414 interactions. We manually took the residues of both X and Y to see which were interacting. Using CLUSTALW we found out the conserved sequences across different strains of bacteria.

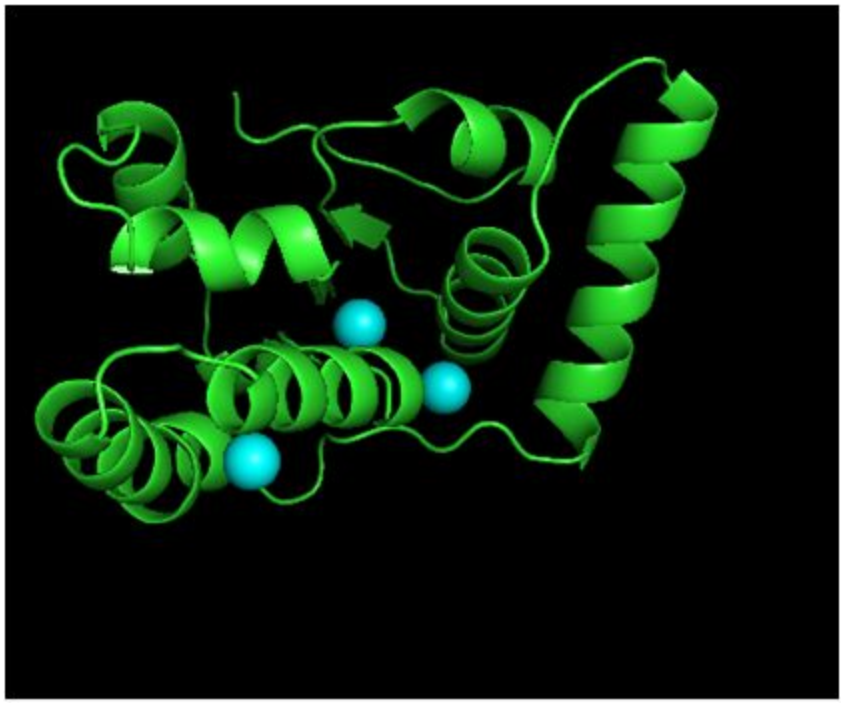


Figure 2 : Antitoxin with water molecules

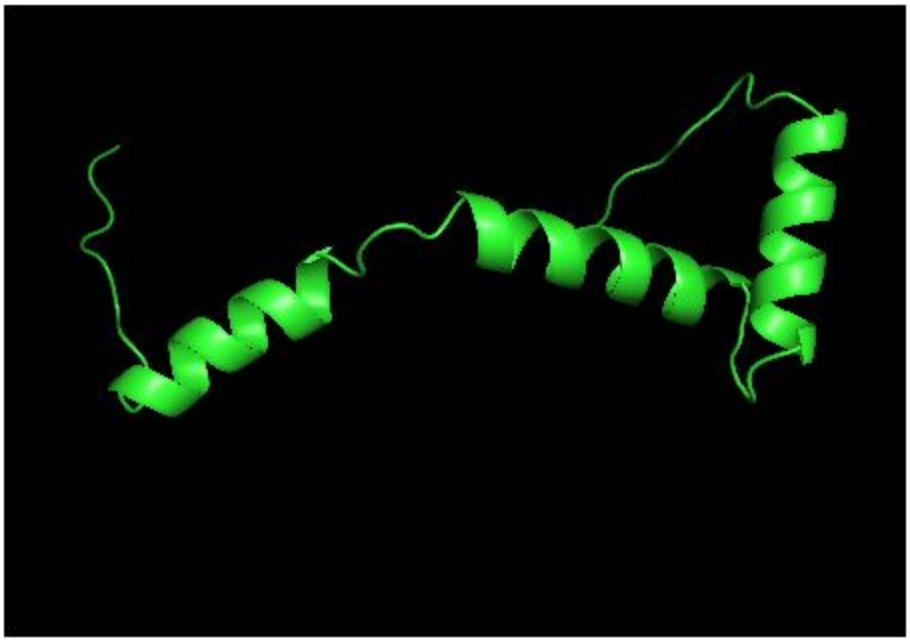


Figure 3 : Toxin molecule

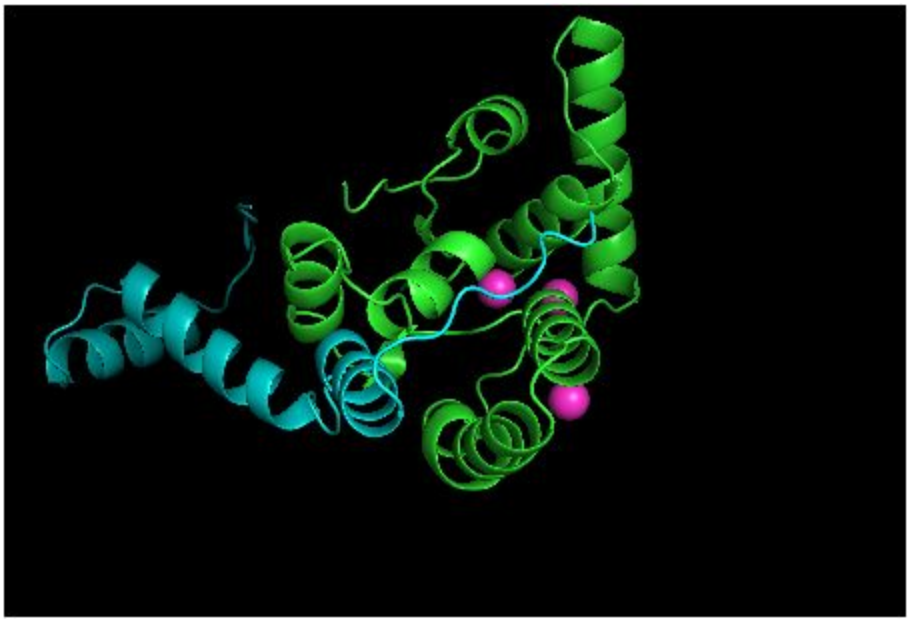


Figure 4 : Antitoxin-toxin unit after PatchDock

Hence we were able to find the conserved residues in both toxin and antitoxin strains. This project focuses on predicting the structure, critical residues, and interacting sites of antitoxin and toxin pairs of *Neisseria Lactamica* and use Molecular Dynamics Simulation to study the dynamics of these proteins. We want

to investigate more to find the structure of antitoxin toxin pairs and figure out the interaction between proteins.

```
1. CD1     31 LEU    4 ----- CD2    364 LEU   48        3.162
2. CB      45 THR    6 ----- NH1    504 ARG   68        4.315
3. OG1     46 THR    6 ----- NH1    504 ARG   68        3.785
4. CA      64 ILE    9 ----- CD1    389 LEU   52        4.172
5. C       65 ILE    9 ----- CD1    389 LEU   52        4.054
6. O       66 ILE    9 ----- CB     387 LEU   52        4.433
7. O       66 ILE    9 ----- CG     388 LEU   52        4.240
8. O       66 ILE    9 ----- CD1    389 LEU   52        3.183
9. CB      67 ILE    9 ----- CD1    389 LEU   52        4.341
10. CG1     68 ILE    9 ----- CD1    389 LEU   52        3.293
11. CD1     70 ILE    9 ----- CD1    389 LEU   52        4.230
12. CA      87 PRO   12 ----- N      391 ALA   53        4.385
13. CA      87 PRO   12 ----- CA     392 ALA   53        4.446
14. C       88 PRO   12 ----- C      385 LEU   52        4.169
15. C       88 PRO   12 ----- O      386 LEU   52        3.916
16. C       88 PRO   12 ----- N      391 ALA   53        4.182
17. C       88 PRO   12 ----- CA     392 ALA   53        4.107
18. O       89 PRO   12 ----- C      385 LEU   52        3.833
19. O       89 PRO   12 ----- O      386 LEU   52        3.463
20. O       89 PRO   12 ----- N      391 ALA   53        3.696
21. O       89 PRO   12 ----- CA     392 ALA   53        3.270
22. O       89 PRO   12 ----- C      393 ALA   53        4.182
23. O       89 PRO   12 ----- O      394 ALA   53        4.134
24. O       89 PRO   12 ----- CB     395 ALA   53        4.121
25. O       89 PRO   12 ----- CA     411 GLY   56        4.199
26. CB      90 PRO   12 ----- C      367 GLY   49        4.396
27. CB      90 PRO   12 ----- O      368 GLY   49        3.386
28. CB      90 PRO   12 ----- CA     384 LEU   52        4.285
29. CB      90 PRO   12 ----- C      385 LEU   52        3.775
30. CB      90 PRO   12 ----- O      386 LEU   52        4.277
```

Figure 5 : A small sampling from a C program to find interactions less than 4.5 Armstrongs

Total interactions found =414

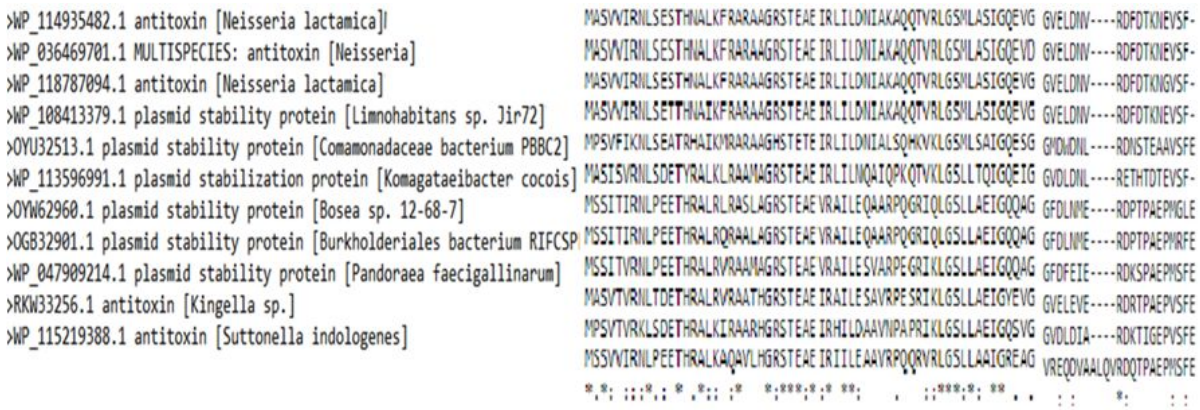


Figure 6 : CLUSTERW Analysis

Molecular Dynamics Simulations :

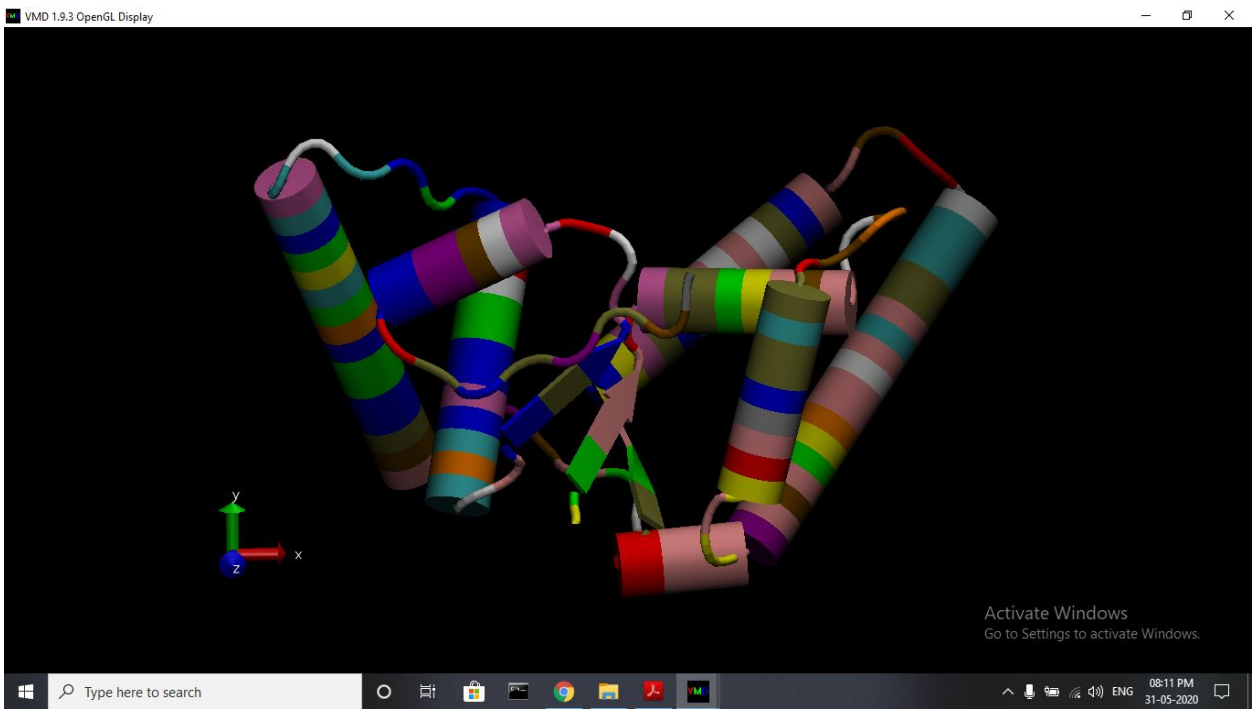


Figure 7 : Neisseria Lactamica
Different residues are of different colour

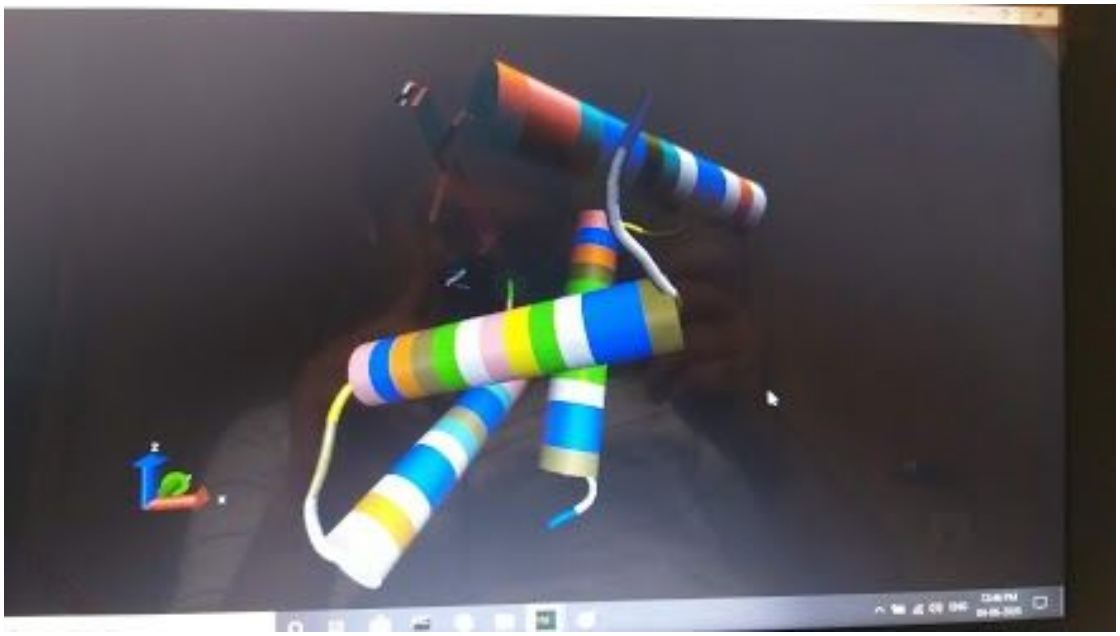
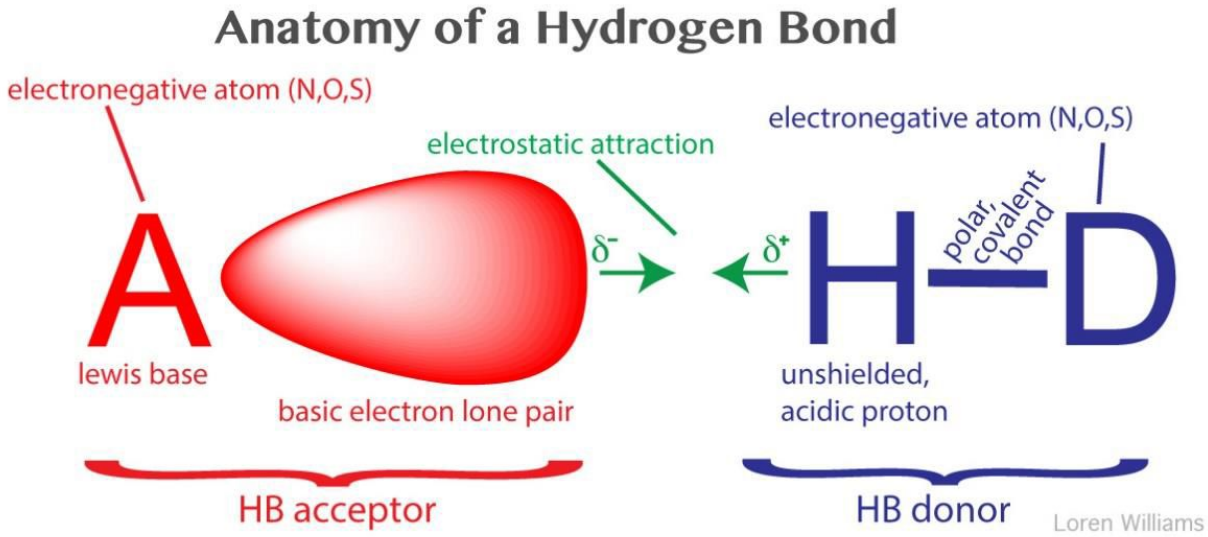
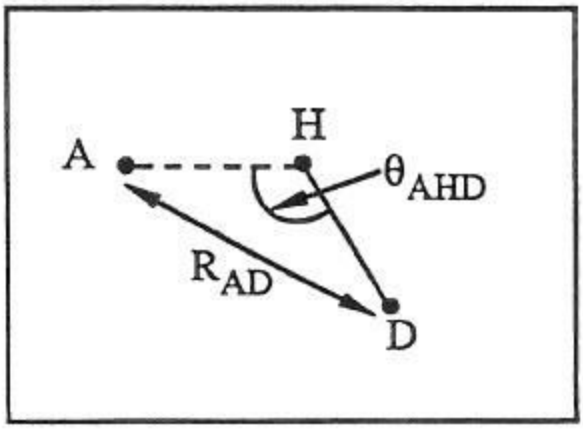


Figure 8 : Toxin molecule



Hydrogen bond Anatomy



Hydrogen bond

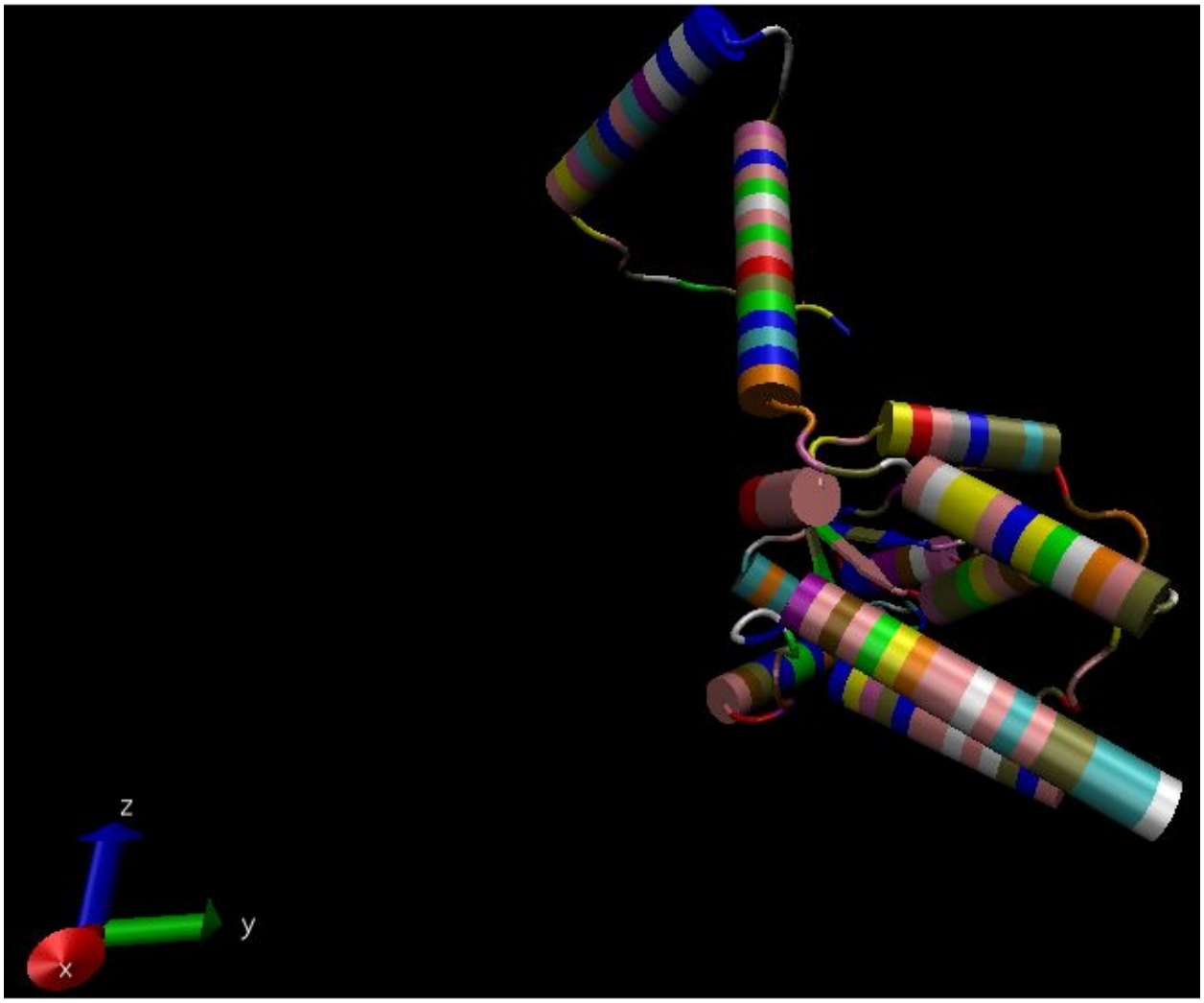


Figure 9 : Resultant complex with residues coloured

With the help of NAMD and VMD software we are able to visualize molecules better. After preparation minimization and solvation we were able to reach the last step of simulation. However during running the simulation we faced a lot of problems. Energy after every step of simulation was increasing for the complex which led to Constraint Failure in RATTLE Algorithm. Even after changing time step and increasing the size of water box we were unable to run the simulation on my personal computer. Hence this report reflects my work but I do not have RMSD statistics associated with the toxin, antitoxin and the complex as I had hoped.

Conclusion and Future Work

Most of the work done by me was computational in nature. Hence it has to be verified by experimentation. Only then can we surely say how the antitoxin - toxin pair looks like and what their interacting residues are. Hence experimentation is something that must be done in future so as to verify our findings.

The Molecular Dynamics Simulations are done with a PC and hence are very primitive. I was able to get some simulations done with very limited resources. Eventually with the help of GPU, one can work and get better results.

Few limitations of MD simulations are :

1. Timescales : Simulations require short time steps for numerical stability – 1 time step ≈ 2 fs (2×10–15 s)
   • Structural changes in proteins can take nanoseconds (10–9 s), microseconds (10–6 s), milliseconds (10–3 s), or longer – Millions to trillions of sequential time steps for nanosecond to millisecond events (and even more for slower ones)
   • Until recently, simulations of 1 microsecond were rare
   • Advances in computer power have enabled microsecond simulations, but simulation timescales remain a challenge
   • Enabling longer-timescale simulations is an active research area, involving: – Algorithmic improvements – Parallel computing – Hardware: GPUs, specialized hardware

2. Force field accuracy
   • Molecular mechanics force fields are inherently approximations
   • They have improved substantially over the last decade, but many limitations remain
   • In practice, one needs some experience to know what to trust in a

simulation.

3. Covalent bonds cannot break or form during (standard) MD simulations
   • Once a protein is created, most of its covalent bonds do not break or form during typical function.
   • A few covalent bonds do form and break more frequently (in real life): – Disulfide bonds between cysteines – Acidic or basic amino acid residues can lose or gain a hydrogen (i.e., a proton)

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